INCH-POUND
FF-S-200D
30 August 2024
SUPERSEDING
FF-S-200C
19 May 2021

# FEDERAL SPECIFICATION

# SETSCREWS: HEXAGON SOCKET AND SPLINE SOCKET, HEADLESS

The General Services Administration has authorized the use of this federal specification by all federal agencies, some states and localities.

## 1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hexagon socket and spline socket headless setscrews.

1.2 Classification.

1.2.1 Types. Setscrews shall be of the following types, as specified (see 6.2)

Type I - Hexagon socket, headless.
Type II - Spline socket, headless.

1.2.2 Styles. Setscrews shall be of the following styles, as specified (see 6.2):

Style 1 - Flat point.
Style 2 - Cone point.
Style 3 - Oval point.
Style 4 - Cup point.
Style 5 - Full-dog point.
Style 6 - Half-dog point.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5305

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1.2.3 Size. Setscrews shall be classified by size in accordance with the basic major diameter of the thread (see 6.2).

1.2.4 Length. Setscrews shall be of the length specified in 6.2.

1.2.5 Part identification number system. A part identification number system shall be used to identify each fastener specified in this document. Part identification may use the numbering system called out on a non-government or military standard (e.g. AN565B1032H12 or MS51476-01 or MS51963-1), but they must specify which part number identification system is being used.

## 2 APPLICABLE DOCUMENTS

2.1 Government publications.

2.1.1 Specifications, standards and handbooks. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATIONS

| | | |
|---|---|---|
| QQ-N-281 | - | Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings, and Structural and Special Shaped Sections |

MILITARY SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-495 | - | Finish, Chemical, Black, for Copper Alloys |
| MIL-DTL-13924 | - | Coating, Inorganic, Black, for Ferrous Metals |
| MIL-DTL-18240 | - | Fastener Element, Self-Locking, Threaded Fastener, 250 Degrees F Maximum |

MILITARY STANDARDS

| | | |
|---|---|---|
| MS51033 | - | Setscrew-Hexagon Socket, Flat Point, Brass, Black Oxide, UNF-3A |
| MS51040 | - | Setscrew-Hexagon Socket, Cone Point, Brass, Black Oxide, UNF-3A |
| MS51476 | - | Setscrew, Hexagon Socket, Cup Point, Alloy Steel, UNC-3A |
| MS51477 | - | Setscrew, Hexagon Socket, Cup Point, Alloy Steel, UNF-3A |

(Copies of military specifications and standards are available for download from the ASSIST database at https://quicksearch.dla.mil.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.



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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

| | | |
|---|---|---|
| ASME B1.1 | - | Unified Inch Screw Threads (UN and UNR Thread Form) |
| ASME B18.3 | - | Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series) |

(Private sector and civilian agencies may purchase copies of these voluntary standards from the American Society of Mechanical Engineers, Two Park Avenue, New York, NY 10016-5990, or through their website at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

| | | |
|---|---|---|
| ASQ Z1.4 | - | Sampling Procedures and Tables for Inspection by Attributes |

(Private sector and civilian agencies may purchase copies of these voluntary standards from the American Society of Quality, P.O. Box 3005, Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, WI 53203, or through their website at www.asq.org.)

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM A342/A342M | - | Standard Test Methods for Permeability of Weakly Magnetic Materials |
| ASTM B36/B36M | - | Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar |
| ASTM B134/B134M | - | Standard Specification for Brass Wire |
| ASTM B633 | - | Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel |
| ASTM D3951 | - | Standard Practice for Commercial Packaging |
| ASTM E18 | - | Standard Test Methods for Rockwell Hardness of Metallic Materials |
| ASTM E1282 | - | Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials |
| ASTM F1941/F1941M | - | Standard Specification for Electrodeposited Coatings on Mechanical Fasteners, Inch and Metric |

(Private sector and civilian agencies may purchase copies of these voluntary standards from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA, 19428-2959, or through their website at www.astm.org.)



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SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE AIR4127 | - | Steel: Chemical composition and Hardenability |
| SAE AMS2461 | - | Plating, Zinc-Nickel Alloy (12 to 16% Ni) |
| SAE AMS2700 | - | Passivation of Corrosion Resistant Steels |
| SAE AMS-QQ-N-290 | - | Nickel Plating (Electrodeposited) |
| SAE AMS-QQ-P-416 | - | Plating, Cadmium (Electrodeposited) |

(Private sector and civilian agencies may purchase copies of these voluntary standards from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096, or through their website at www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3 REQUIREMENTS

3.1 Material. Setscrews shall be made of one of the following material as specified (see 6.2).

3.1.1 Alloy Steel. Alloy steel shall be AISI 4032, 4037, 40L37, 4137, 4140, 41L40, 8630, 8740, in accordance with SAE AIR4127.

3.1.2 Corrosion-resisting steel.

3.1.2.1 Austenitic. Austenitic corrosion-resisting steel shall be of the 300 Series in accordance with SAE AIR4127 or alloys 316/316L. For superior corrosion resistance in the marine environment, alloys 316 or 316L shall be specified (see 6.2).

3.1.2.1.1 Magnetic permeability. Austenitic corrosion-resisting steel setscrews shall have a magnetic permeability of 2.0 max (air = 1.0) for a field strength of H = 200 oersteds when using a permeability indicator in accordance with ASTM A342/A342M.

3.1.2.2 Martensitic. Martensitic corrosion-resisting steel shall be Steel Nos. 410 or 416 in accordance with SAE AIR4127.

3.1.3 Brass. Brass shall be in accordance with alloy no. C26000 or C26800 of ASTM B36/B36M or alloy no. C26000, C27000 or C27400 of ASTM B134/B134M.

3.1.4 Nickel copper. Nickel copper shall be in accordance with QQ-N-281, Class A (Form 1: Bar and Rod or Form 7: Wire), or Class B (Form 1: Bar and Bar or Form 2: Wire).

3.2 Protective finishes. Unless otherwise specified in the contract, setscrews shall be furnished uncoated. When specified (see 6.2), protective finishes shall be as follows.



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3.2.1 Plating/coating. When plating is required for setscrews of 48 pitch or finer, the thickness may be reduced to .0001 inch minimum thickness. Hydrogen embrittlement relief treatment is mandatory for plating (see 3.2.4). When specified in the procurement documents, plate/coat the fastener in accordance with the following:

a. Cadmium – SAE AMS-QQ-P-416/1, Type II, Class 3
b. Zinc-Nickel – SAE AMS2461, Type II, Grade B, Class 3/3
c. Zinc – ASTM B633 Fe/Zn 12 Type II/2
d. Zinc (hexavalent chromium-free passivation) – ASTM F1941/F1941M Fe/Zn 12 AN
e. Nickel – SAE-AMS-QQ-N-290

/1 Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per SAE AMS2461 is a suggested alternate with design authority approval.
/2 Type II passivation per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc with hexavalent chromium-free passivation per ASTM F1941/F1941M Fe/Zn 12 is a suggested alternate with design authority approval.
/3 SAE AMS2461 is an acceptable substitute for MIL-PRF-32647, Type K16 or Type K20 (included in Revision C of FF-S-200) with design authority approval.

3.2.2 Passivation. Corrosion-resisting steel setscrews shall be passivated in accordance with SAE AMS2700.

3.2.2.1 Passivation hex chrome-free. Corrosion-resisting steel setscrews shall be passivated in accordance with SAE AMS2700, Method 1, Type 6, 7 or 8, or Method 2; no post-treatment.

3.2.3 Black oxide.

3.2.3.1 Brass. Brass setscrews shall be treated with a black chemical in accordance with MIL-DTL-495.

3.2.3.2 Corrosion-resisting steel. Corrosion-resisting steel setscrews shall be black oxide coated in accordance with MIL-DTL-13924, Class 4.

3.2.4 Hydrogen embrittlement. Hydrogen embrittlement relief for plated alloy steel setscrews shall be provided in accordance with the applicable plating specification.

3.3 Hardness. Alloy steel setscrews shall be hardened by quenching in oil from the hardening temperature and tempered to a hardness of Rockwell 45-53 HRC. Martensitic corrosion-resisting steel setscrews shall be hardened by heat treatment to Rockwell 35-45 HRC. Austenitic corrosion-resisting steel setscrews shall have a hardness of Rockwell 96 HRB – 33 HRC and brass alloy setscrews shall have a hardness of Rockwell 55 – 80 HRF.

3.3.1 Decarburization. The complete end partial decarburization of the threads of heat-treated setscrews shall not exceed the amounts shown in Figure 1 or the amount for such other class as may be specified in the contract or order. A carbon content throughout the threads equivalent to that of the core is desirable.



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3.4 Design.

3.4.1 Dimensions. Dimensions and tolerances of setscrews shall be in accordance with ASME B18.3. For acceptable designs of cup points for military use, see applicable Military Standards below:

MS51033
MS51040
MS51476
MS51477

3.4.2 Threads. Unless otherwise specified (see 6.2), the threads shall be class 3A in both UNC and UNF series in accordance with ASME B1.1.

3.4.3 Self-locking element. When specified (see 6.2) non-metallic, self-locking elements shall be in accordance with MIL-DTL-18240.

3.5 Workmanship. Setscrews shall be free from burrs, seams, laps, loose scale, irregular surfaces and any other defects which may affect serviceability. Unless otherwise specified (see 6.2), broach petal (chips) are acceptable provided they will not affect serviceability.

## 4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use their own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. Inspection records of the examination and tests with itemized results shall be kept complete at the manufacturer's facility, available to the Government throughout the duration of the contract, or a minimum of two years, whichever is longer. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet requirements of sections 3, 4 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to accept defective material.



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4.2 Quality conformance inspection.

4.2.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in application referenced documents.

4.2.2 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and marking comply with the requirements in Section 5 of this specification.

4.2.3 Lot. A Lot shall consist of all setscrews of the same type, style, mill heat of material, protective finish, size, length and threads produced and manufactured at the same time by the same manufacturer's facility. All setscrews in a lot shall be offered for acceptance at the same time.

4.2.4 Sampling for examination. A random sample of setscrews shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level II. The Acceptable Quality Limit (AQL) shall be as specified in Table I. Unless otherwise specified, the lot size and inspection level in Table II-A of ASQ Z1.4 shall be used to establish the sample size. However, the acceptance number shall be zero.

4.2.5 Sampling for hardness and decarburization tests. A random sample of setscrews shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level S-1, with an AQL of 1.0 percent defective for hardness test. Three (3) of these hardness test setscrews shall be subjected to decarburization test as specified in 4.3.2.

4.2.6 Sampling for protective finish tests. Sampling for tests of protective finishes shall be in accordance with the applicable specifications in 3.2.

4.2.7 Sampling for chemical analysis. When specified (see 6.2), a random sample of setscrews shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level S-1, with an AQL of 1.5 percent defective. Samples previously selected for examination or for the hardness and decarburization tests may be used.

4.2.8 Examination. Each setscrew taken as specified in 4.2.4 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table I. Any setscrew in the sample containing one or more defects shall be rejected and if the number of defective setscrews in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Short setscrews passing through the no-go side of the gauge shall not be cause for rejection, providing the screws have less than six threads (including the start thread) and the major diameter shall not be undersize by more than 0.003 inch.



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TABLE I. Classification of defects.

| Class of Defect | Type of Defect | Inspection Method |
|---|---|---|
| Critical | None defined | |
| | Major AQL = 2.5 | |
| 101 | Thread | SIE /1 |
| 102 | Socket dimensions | SIE |
| | Minor AQL = 4.0 | |
| 201 | Magnetic permeability (3.1.2.1.1) | SIE |
| 202 | Applicable point dimensions | SIE |
| 203 | Setscrew length | SIE |
| 204 | Chamfer angle | SIE |
| 205 | Eccentricity | SIE |
| 206 | Workmanship (3.5) | Visual |

/1 SIE = Standard Inspection Equipment

4.3 Test methods.

4.3.1 Hardness. Each setscrew taken as specified in 4.2.5 shall be subjected to a hardness test conducted in accordance with ASTM E18.

a. Setscrews having a length of 2 diameters or longer shall be tested on a transverse section through the threads at a point one-quarter of the nominal diameter from the axis and one nominal diameter from the point.
b. Setscrews having a length less than 2 diameters shall be tested on a transverse section through the threads at a point one-quarter of the nominal diameter from the axis and located at the first full thread from the point.
c. Setscrews having nominal diameters of 1/4” and larger shall be tested using the Rockwell C Scale. For sizes smaller than 1/4” the Rockwell 30N Scale shall be used.

4.3.2 Decarburization. Each setscrew taken as specified in 4.2.5 shall be subjected to a decarburization test conducted as follows: The threaded part of the setscrew shall be sectioned longitudinally through the axis and polished so that rounding of the thread edges is held to a minimum. The section shall be etched with the usual metallographic etchants (such as Nital) and examined with a metallurgical microscope. Measurements shall be made from the crest of the thread to the end of the decarburized zone. This distance shall not exceed that indicated in Figure 1. Measurements shall be made with a fixed-scale micrometer eyepiece. This test does not apply to austenitic corrosion-resisting steel and brass alloy.

4.3.3 Protective finish. Protective finish tests shall be conducted in accordance with the applicable specifications in 3.2.



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4.3.4 Chemical analysis. Each setscrew selected as specified in 4.2.7 shall be subjected to a chemical analysis conducted in accordance with ASTM E1282 to verify conformance to the applicable material specification. A steel mill certification will be acceptable in lieu of this test.

4.3.5 Hydrogen embrittlement. The contractor shall furnish the Government certification that cadmium plated, zinc plated or zinc-nickel plated alloy steel setscrews have been subjected to the embrittlement relief treatment specified in 3.2.4.

## 5 PACKAGING

5.1 Packaging. The requirements for packaging shall be in accordance with ASTM D3951, Standard Practice for Commercial Packaging, unless detailed requirements are specified in the contract.

## 6 NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Setscrews covered by this specification are intended to be used to locate machine parts in relation to each other, either permanently or temporarily. It is recommended that cup point setscrews, style 4, be used wherever possible. For hardened shafts or when the cutting in of the cup point is objectionable, styles 1 or 6 should be used. It is recommended that styles 2, 3 or 5 be used only when it is impractical to use the other styles.

6.2 Acquisition data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, style, size, length, thread series and class (1.2).
c. Material (3.1).
d. Protective finish, if required (3.2).
e. Self-locking feature, if required (3.4.3).
f. Chemical analysis, when required (4.2.7).
g. Broach petals removal, if required (3.5).
h. Packaging requirements (see 5.1)

6.3 Military procurement. Items procured under this specification for military use are to be limited to the variety shown on the applicable Military Standards. Personnel of Military departments are requested to refer to these documents for guidance.



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6.4 Supersession data. Cross reference between the types and styles of setscrews covered in part by this specification and FF-S-210, and the types and styles of the superseded FF-S-103, dated October 7, 1958, is as follows:

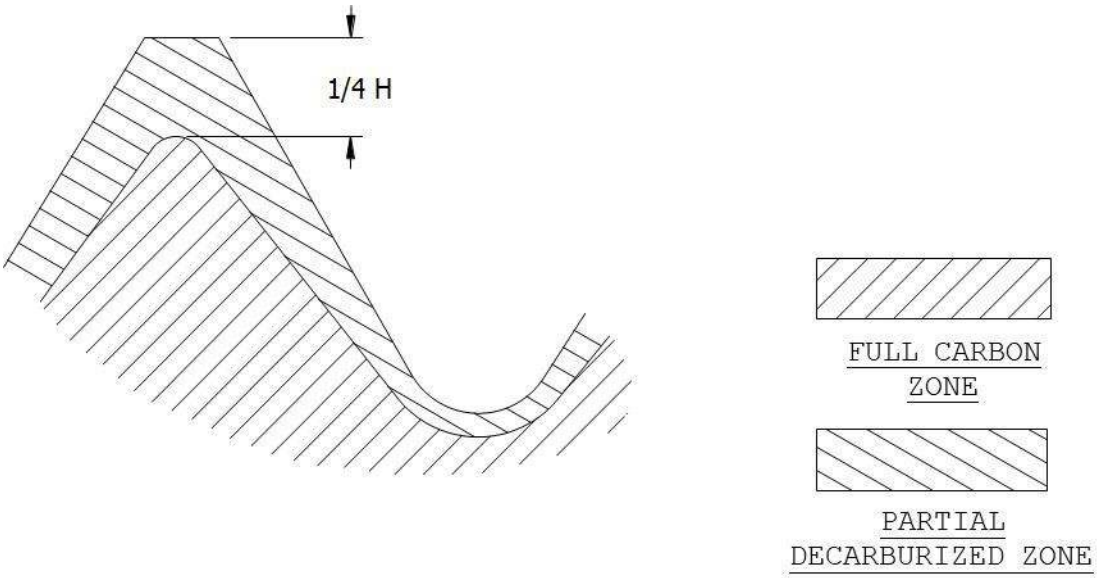
TABLE II. Cross reference data.

| FF-S-1O3 | FF-S-210 | FF-S-200 |
|---|---|---|
| Types | Types | Types |
| Type I - Square Head | Type I - Square Head | --- |
| Type II - Slotted Headless | Type II - Slotted Headless | --- |
| Type III - Hexagon Socket, Headless | --- | Type I - Hexagon Socket, Headless |
| Type IV - Fluted Socket, Headless | --- | Type II - Spline Socket, Headless |
| Styles | Styles | Styles |
| Style 1 - Flat point | Style 1 - Flat point | Style 1 - Flat point |
| Style 2 - Cone point | Style 2 - Cone point | Style 2 - Cone point |
| Style 3 - Oval point | Style 3 - Oval point | Style 3 - Oval point |
| Style 4 - Cup point | Style 4 - Cup point | Style 4 - Cup point |
| Style 5 - Full-dog point | Style 5 - Full-dog point | Style 5 - Full-dog point |
| Style 6 - Half-dog point | Style 6 - Half-dog point | Style 6 - Half-dog point |

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.



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| Thds per inch | 1/4 H |
|---|---|
| 80 | .002 |
| 72 | .002 |
| 64 | .003 |
| 56 | .003 |
| 48 | .003 |
| 44 | .004 |
| 40 | .004 |
| 36 | .005 |
| 32 | .005 |
| 28 | .006 |
| 24 | .007 |
| 20 | .008 |
| 18 | .009 |
| 16 | .010 |
| 14 | .012 |
| 13 | .012 |
| 12 | .014 |
| 11 | .015 |
| 10 | .016 |
| 9 | .018 |
| 8 | .020 |
| 7 | .023 |
| 6 | .027 |

(1/4 H - l/4 Basic Thread Depth)

Figure 1. Decarburization limits.



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DLA - GS

Review activities:
Army - AV, CR4, EA, MI
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Preparing activity:
Army - AT

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